
FALCON's Response to RFQ ThermoFisher : Supply of PTL System

December 31, 2024

Doc. No. - Sales/FR/PT/007/R0



Kind Attention –

Mr. Aniket

Offer Ref: F24-00295; 31.12.2024.

Subject – Techno-Commercial Offer for PTL System

Dear Team,

We are pleased to submit our Techno-Commercial Offer in response to your requirement. We have taken this opportunity to also include adjustments discussed during online meetings.

We would like to highlight the fact that Falcon Autotech, as a prime contractor for this project, is a global leader in providing intra-logistics automation solutions. Falcon has installed over **150 sorters worldwide** with capacity ranging from **800 to 60,000 PPH** and has an excellent track record in designing, manufacturing, and installing parcel sortation systems.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

Company Details

Company Name : Falcon Autotech Pvt. Ltd.

Website : www.falconautotech.com

LinkedID : <https://www.linkedin.com/company/falconautotech/mycompany/>

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Table of Contents

1. Glossary	4
2. Company Profile.....	5
3. Falcon's Experience and Achievements in Sortation Space Globally	9
4. Proposed System Description.....	11
4.1 Objective	11
4.2 Summary of the System	11
5. Layout View.....	12
6. System Component.....	13
6.1 Pick/Put to Light Module	13
6.2 Idler Roller Conveyor	14
6.3 Push Back Racks With Bins.....	15
6.4 Pallet Frame	15
6.5 System HMI	16
6.6 Handheld Scanner 1D	16
7. Proposed System Technical Details/BOM	17
7.1 Mechanical equipment	17
7.2 Electrical Equipment	17
7.3 Control System.....	17
8. Falcon's WCS CONTROLIT.....	18
9. Commercial Terms.....	19
9.1 Price Sheet.....	19
10. Dispatch Timelines	20
11. Payment Terms	20
12. Warranty Period	20
13. Exclusions	21

1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour
3	IT	Information Technology
4	BOQ	Bill Of Quantity
5	I/O	Input/ Output
6	PDP	Power Distribution Panel
7	PC	Personal Computer
8	UPS	Uninterrupted Power Supply

2. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0

Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5
Product Lines
 15+
Countries with Live Installations
 1800+
Total Installed Systems Globally
 1Million+
Sorts Per Hour
 600+
Employees

Long-standing Relationships with industry leaders

News & Articles

FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

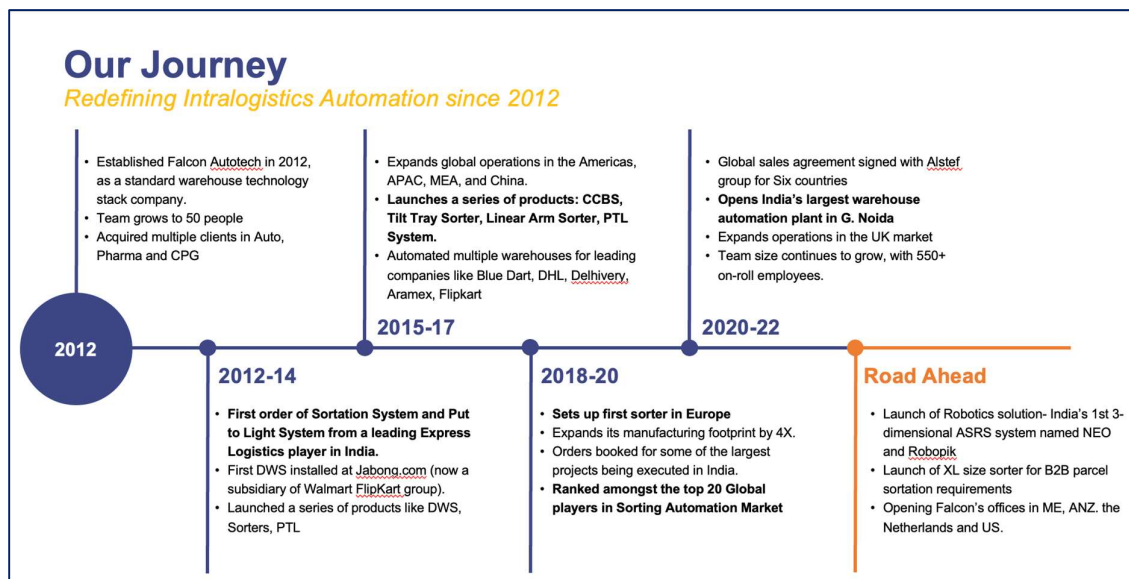
FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation companies; our vision is to become the top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

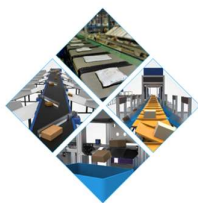


As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. To be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and delivering world class projects in return.



Product and Solutions

With 100% focused on Parcels, Packages, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- Popup Sorter
- Sweep Sorter
- Pusher Sorter



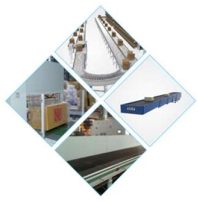
PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



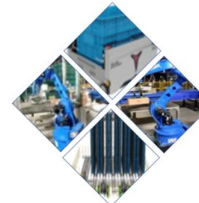
DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
 - R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
 - Mini (600MM)
 - Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

Powered by

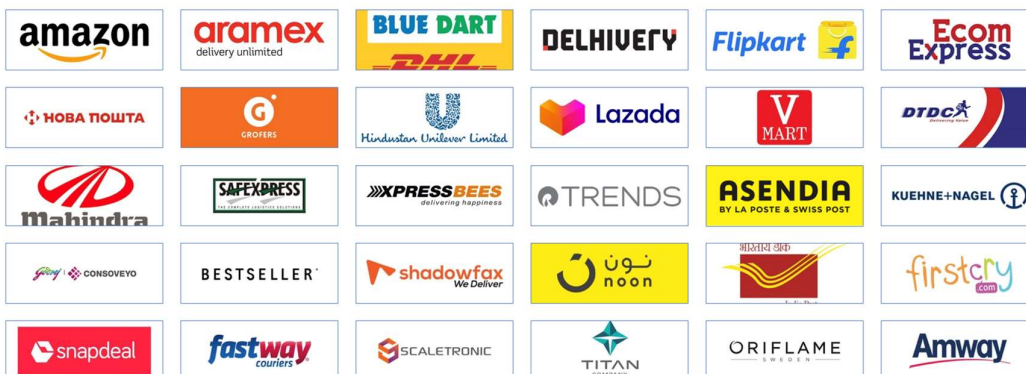


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of the above product lines for effective materials handling, sortation, and movement. The process is controlled in real-time by our in-house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions, and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long-standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

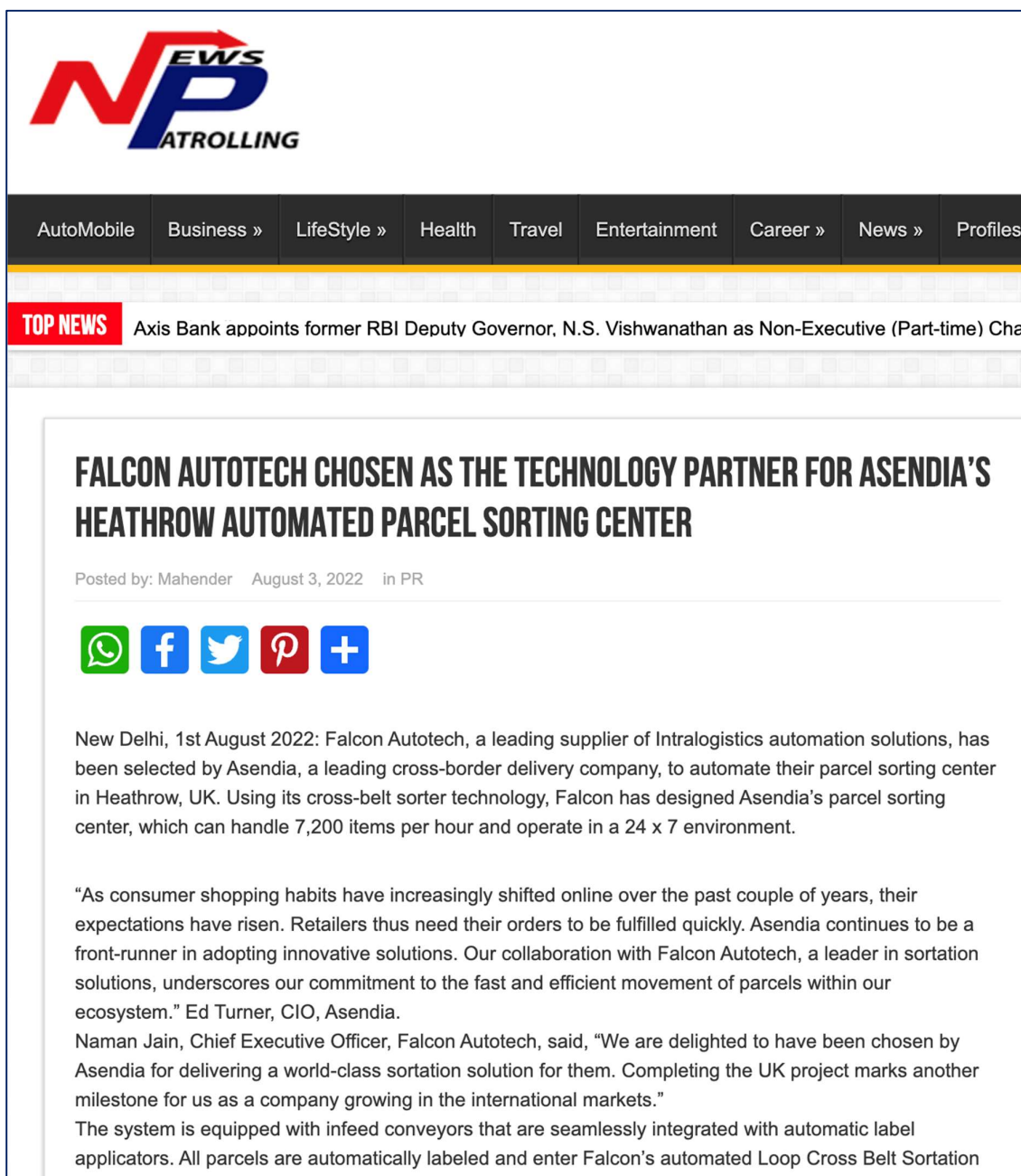


With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.



3. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



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TOP NEWS Axis Bank appoints former RBI Deputy Governor, N.S. Vishwanathan as Non-Executive (Part-time) Chair

FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation

POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

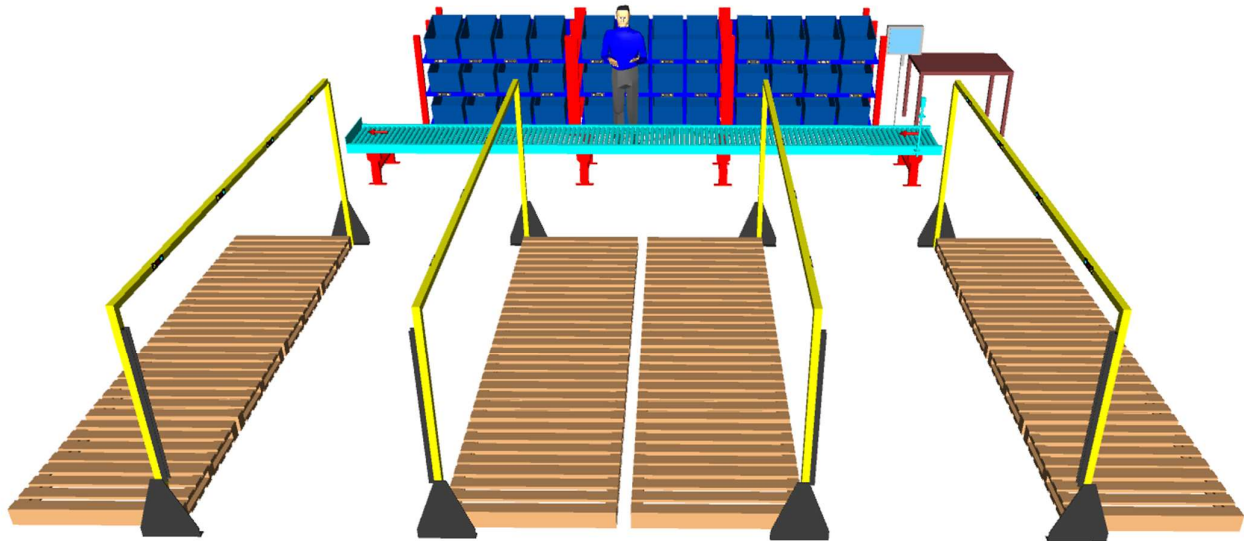
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

4. Proposed System Description

4.1 Objective

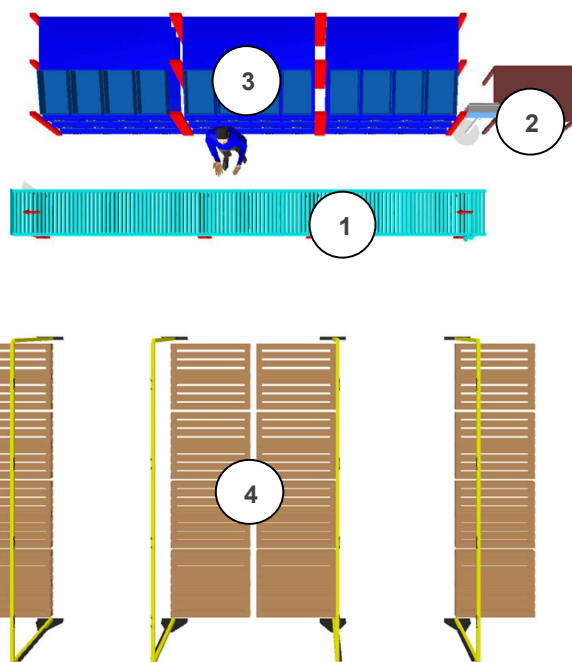
The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing and acceptance testing of the complete PTL system to sort products as per specifications shared by ThermoFisher.



4.2 Summary of the System

1. **Infeed:** SKU Cartons are fed into the system through 1 Nos Infeed Idler roller lines running throughout the rack Position.
2. **Picking:** Carton Is scanned & PTL Light Glows indicating the put position Along with number of SKU.

5. Layout View



Legend

- 1 – Idler Conveyor
- 2 – System HMI
- 3 – Pushback Rack & Bins (With PTL Lights)
- 4 – Pallet Frame (With PTL Lights)

6. System Component

6.1 Pick/Put to Light Module

PTL system is falcons in house developed modular light hardware modules and a connected software platform to ensure a flexible, robust, seamless, and easy to setup PTL solution which is fully integrated with Client's WMS/ERP Systems.

The Software platform is powered by a deep artificial intelligence engine that continually mines the data generated by operations and keeps on altering the PTL configuration to increase the system productivity and give you an extra competitive edge.

The System involves a collection of physical sorting locations representing either a customer order/ address location or any other logical sorting group. Each physical location is mapped to a light module which glows on the event of scan of a product for that sorting location.

It is a paperless technology that provides high throughput with a single operator with high accuracy.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Dimensions	Mm	Around 165 X 50
PTL location	Nos	As per Layout requirement
LED Display	Type	3 Digit Alpha numeric Rotating Display
LED Size	Mm	~ 25 X 20
LED Indicator	yes/no	YES
Acknowledgment Button	yes/no	YES
Function keys	yes/no	4 Keys
Power Requirements	Type	3 Phase UPS Power
Communication	Type	TCP/IP and Serial
Operating temperature	Deg.	45 Degrees
Software Interface	Type	Over APIs or other methods
Type of mount	Type	Clip On Type on Rails

6.2 Idler Roller Conveyor

Falcon's Roller conveyors require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's idler roller conveyor:

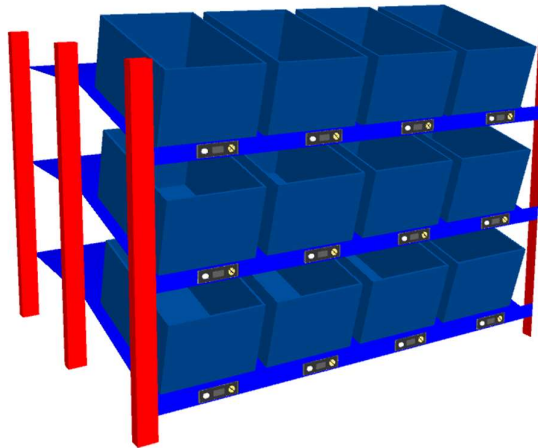
- Low Noise
- Maximum uptime.
- Minimal maintenance
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	600
Conveyor Height	mm	As per Layout requirement
Load capacity per unit length of Conveyor	kg/m	50
Roller Span	mm	approx. 600
Roller Diameter	mm	approx. 50
Pitch of Rollers	mm	75
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions
Type of Guides	Type	Yes
End Stopper	yes/no	Yes

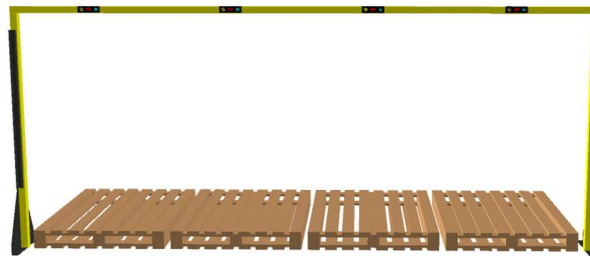
6.3 Push Back Racks With Bins

A push-back rack is a type of storage system that optimize space and improve accessibility. It consists of a series of inclined rails or shelves where bins or containers are placed. The design allows items to be loaded from one end and pushed back to the rear by newly loaded items, making it easy to access goods without having to move other items out of the way. This system maximizes storage density by utilizing deep storage spaces and is particularly useful for high-density, first-in, last-out inventory management.



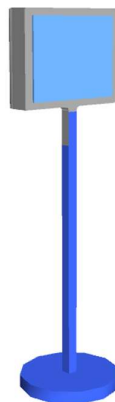
6.4 Pallet Frame

A pallet frame with a put-to-light module is a structured rack that holds pallets positions, while the put-to-light module integrates with the frame to guide the picking process. Each pallet location is equipped with a light that illuminates when it's time to add or retrieve items. This system streamlines the Putting process by providing clear visual cues, reducing errors, and improving speed and accuracy in order fulfillment. It's particularly useful in environments with high order volumes and varied inventory.



6.5 System HMI

HMI screen provides real-time data on system performance, status, and operational parameters. It allows operators to view critical information and offers options to control or adjust system settings. The HMI screen typically features intuitive navigation, graphical displays, and interactive controls, enabling users to efficiently manage and troubleshoot the system.



6.6 Handheld Scanner 1D

A handheld scanner is a portable device used to read and capture data from barcodes or QR codes.. The compact design make them versatile and easy to use in various operational environments.



7. Proposed System Technical Details/BOM

7.1 Mechanical equipment

Pos.	Qty.	Description	Value
1	1	PTL System – 52 Lights	
		Pushback Racks	36 Nos
		PTL Bin	36 Nos
		PTL Pallet Frame	16 Nos
		PTL Module	52 Nos
		PTL Control Box	02 Nos
		HMI	01 Nos
		Handheld Terminal 1D	02 Nos

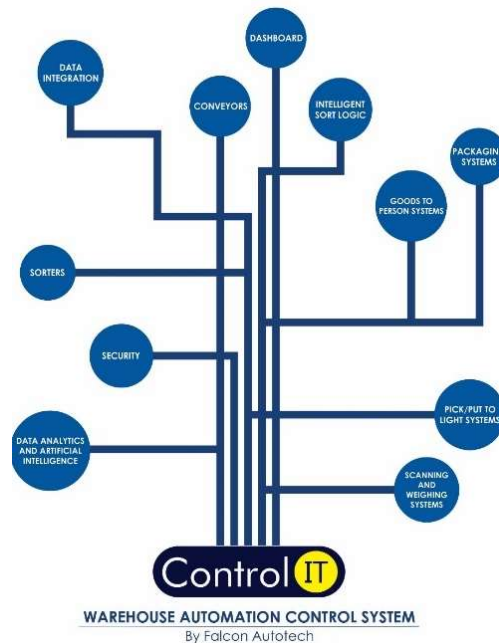
7.2 Electrical Equipment

Pos.	Qty.	Description	Value
1	1	Cabling and Wiring	
		Network Switches	1 Nos
		Field Cabling	1 Nos

7.3 Control System

Components			
1	1	Controls and IT Infrastructure Controls IT	
		Control IT Software Package	
		IT Integration with Main Server	Included

8. Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

9. Commercial Terms

9.1 Price Sheet

Sr. No	Component	Set	UOM	Price(INR)
1	Conveyor Module	1	Set	₹ 1,25,640
2	HMI Display	1	Set	₹ 88,000
3	PTL Bin System	1	Set	₹ 6,00,600
4	PTL Pallet frames	1	Set	₹ 1,79,840
5	Software Packages	1	Set	₹ 4,10,000
6	Installation & Commissioning	1	Set	₹ 49,704
7	Packaging & Forwarding	1	Set	₹ 19,882
8	Project Management and Engineering Charges	1	Set	₹ 29,473
Grand Total				₹ 15,03,139

Sr. No	Component	Set	UOM	Price(INR)
1	Conveyor Module	1	Set	₹ 1,25,640
2	Installation & Commissioning	1	Set	₹ 11,308
3	Packaging & Forwarding	1	Set	₹ 3,141
4	Project Management and Engineering Charges	1	Set	₹ 5,026
Grand Total				₹ 1,45,114

Sr. No	Component	Set	UOM	Price(INR)
1	Pallet PTLs	1	Set	₹ 1,79,840
2	Installation & Commissioning	1	Set	₹ 16,186
3	Packaging & Forwarding	1	Set	₹ 4,496
4	Project Management and Engineering Charges	1	Set	₹ 7,194
Grand Total				₹ 2,07,715

The Total Price is to be understood.

- Ex Works – Greater Noida, India
- Freight charges shall be extra at actual.
- Taxes and duties shall be extra as applicable.
- Including Packing charges
- Including Installation and Commissioning
- Price is valid for 30 days from the date of proposal.

Our Price does not include taxes, and duties, import duties, charges, fees, taxes etc. charged by authorities outside India for deliveries and services. Same shall be borne & paid by client itself.

*Material Unloading & Laydown area is in customer scope

10. Dispatch Timelines

Dispatch within 12-14 weeks from the date of DAP.

11. Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 50% Advance along with PO.

Stage 2: 50% along with 100% Taxes prior to dispatch.

12. Warranty Period

Falcons offered System comes with a standard warranty of 1 year for complete system against any manufacturing defect. Any damage caused to the system by an external source, abnormal use of equipment or any other factor not attributable to Falcon Autotech shall not be covered under warranty.

Post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

13. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- A. Server PC
- B. Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- C. Electrical power supply and wiring to the main control cabinets.
- D. UPS for Controls and Drives
- E. Electrical Power for Installation
- F. Network Cabling up to the Main Server Rack
- G. Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- H. Emergency/Uninterruptable power supply
- I. Fire-alarm and fire protection devices
- J. Traffic and route markings
- K. Laydown Area
- L. Ram protection devices
- M. Cat walks, bridges, maintenance aisles and platforms
- N. Assembly tools like forklifts and hoisting machines.
- O. All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- P. Any Kind of Civil work
- Q. Any adjustment of the Supplier's scope of supply to local rules and regulations
- R. X-Ray machines
- S. Roller cages/ Pallets
- T. Simulation and 3D animation of the system
- U. Interface with other equipment not specified in this offer.
- V. Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- W. The supply and installation of fencing around the different corridors.
- X. Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.